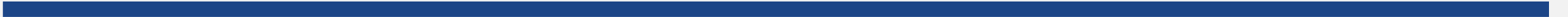


CMP370 Internet of Things

Unit 5- Cloud Computing in IoT



Cloud Computing

- Cloud computing is a transformative computing paradigm that involves delivering applications and services over the Internet.
- Cloud computing involves provisioning of computing, networking and storage resources on demand and providing these resources as metered services to the users, in a "pay as you go" model.
- Cloud computing resources can be provisioned on-demand by the users, without requiring interactions with the cloud service provider.
- The process of provisioning resources is automated. Cloud computing resources can be accessed over the network using standard access mechanisms that provide platform-independent access through the use of heterogeneous client platforms such as workstations, laptops, tablets and smart-phones.

Cloud Computing in IoT

Role of Cloud Computing in IoT

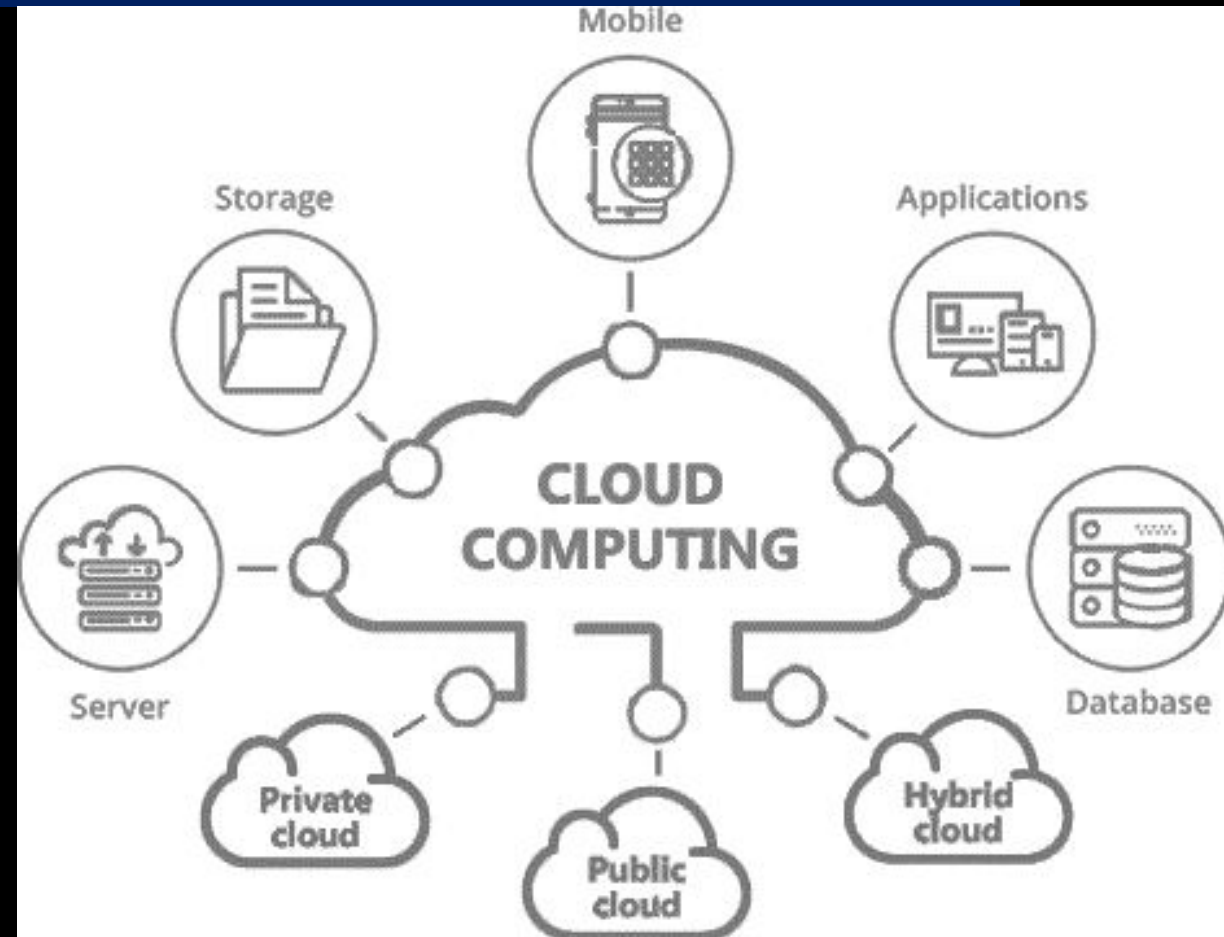
- Cloud computing provides storage, processing, and analytics for IoT data.
- Enables scalability (handling millions of devices), remote access (via dashboards), and data integration (e.g., combining sensor data with AI models).
- Example: Storing temperature data from a smart home in the cloud for trend analysis.

Cloud Computing



Application logic runs
on user's computer

Application logic runs
in the cloud



Cloud Computing

- The computing and storage resources provided by cloud service providers are pooled to serve multiple users using multi-tenancy.
- Multi-tenant aspects of the cloud allow multiple users to be served by the same physical hardware. Users are assigned virtual resources that run on top of the physical resources.
- Cloud computing services are offered to users in different forms
 - a) Infrastructure-as-a-Service (IaaS)
 - b) Platform-as-a-Service (PaaS)
 - c) Software-as-a-Service (SaaS)

Infrastructure-as-a-Service (IaaS)

- IaaS provides the users the ability to provision computing and storage resources. These resources are provided to the users as virtual machine instances and virtual storage.
- Users can start, stop, configure and manage the virtual machine instances and virtual storage.
- Users can deploy operating systems and applications of their choice on the virtual resources provisioned in the cloud.
- The cloud service provider manages the underlying infrastructure. Virtual resources provisioned by the users are billed based on a pay-per-use paradigm.

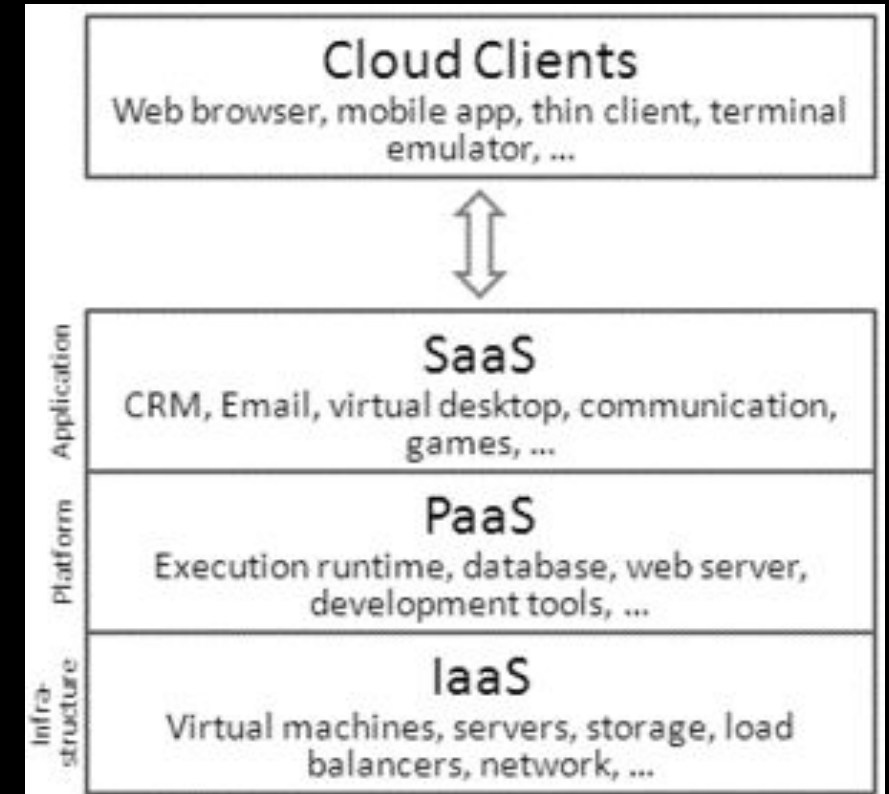
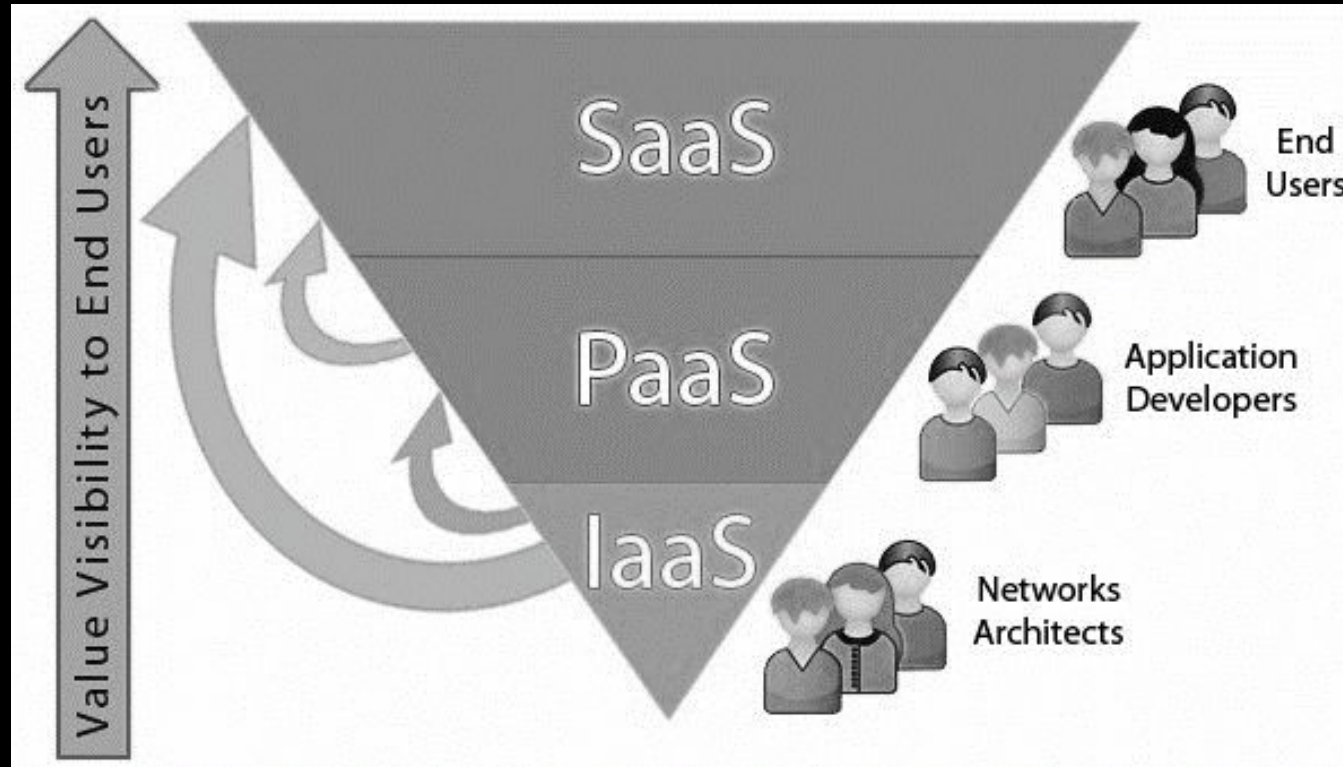
Platform-as-a-Service (PaaS)

- PaaS provides the users the ability to develop and deploy application in the cloud using the development tools, application programming interfaces (APIs), software libraries and services provided by the cloud service provider.
- The cloud service provider manages the underlying cloud infrastructure including servers, network, operating systems and storage.
- The users, themselves, are responsible for developing, deploying, configuring and managing applications on the cloud infrastructure.

Software-as-a-Service (SaaS)

- SaaS provides the users a complete software application or the user interface to the application itself.
- The cloud service provider manages the underlying cloud infrastructure including servers, network, operating systems, Storage and application software, and the user is unaware of the underlying architecture of the cloud.
- Applications are provided to the user through a thin client interface (e.g., a browser), SaaS applications are platform independent and can be accessed from various client devices such as workstations, laptop, tablets and smart-phones, running different operating systems.
- Since the cloud service provider manages both the application and data, the users are able to access the applications from anywhere.

Service Models of Cloud Computing



Artificial intelligence (AI)/Machine learning (ML)

- Owing to the deployment of numerous IoT devices and applications, the complexity and size of data over and beyond the networks have increased significantly.
- The IoT data may or may not be structured and often consists of hidden patterns, which have to be derived through data processing and statistical inferences.
- Modern-day AI/ML based tools have significantly powerful inferencing procedures, which can outperform almost any of the standard statistical methods.
- Additionally, the need for automation in a significant chunk of IoT devices and applications is another compelling reason for the rapid emergence and adoption of AI/ML with IoT.
- AI/ML has been used for extracting information from raw data, be it from agricultural sensors, smart home sensors, or network security analyzers. AI/ML tools are mostly data-driven. However, new methods in these domains are rapidly cropping up; methods which do not always have to rely on voluminous data for generating inferences or predicting trends.

IoT Platform

Various cloud-based Internet of things (IoT) platforms.

1. Google Cloud Platform
2. ThingSpeak
3. Microsoft Azure Cloud
4. Amazon Web Services
5. IBM BlueMix
6. Digital Service Cloud
7. Zetta

Google Cloud Platform

- Google is undoubtedly the most favored IoT stage in light of its worldwide wildlife organization, pay-as-you-use system, big data and investigation apparatus, and accessibility of different cloud administrations like BigQuery, Firebase, Wireless Solutions, Cassandra, Google Cloud Platform, and some more.
- Google foundation is constantly valued as it can assist engineers with coding and test and convey their applications easily. Google foundation is profoundly adaptable and solid. Besides, Google deals with issues for foundation, information stockpiling, and registering power.

Key Features of Google Cloud Platform

- Run on Google's infrastructure
- Scalability
- Compute, storage and services
- Higher performance
- Provide support if required
- Assurance of Google Grade security and compliance for your applications
- Environment safe cloud

ThingSpeak

- ThingSpeak is an open-source stage for IoT application advancement.
- It is equipped for coordinating your information with an assortment of outsider stages, frameworks, and innovations, including other driving IoT stages. This stage gathers and sends the gadget information gathered from the cloud for additional investigation.
- Post examination empowers the IoT framework and application. Later, the outcomes can be seen through the representation instruments.

Key Features of ThingSpeak

- It provides real time collection of data storage
- Data analytics and Visualization using MATLAB
- Device Communication
- Open API support
- Provides Geolocation data
- Facilitates plugins

Microsoft Azure Cloud

- Microsoft Azure Intelligent System Service shapes a firm stage and administrations. By getting together, putting away, and preparing information, it constructs Internet of things frameworks and applications.
- The shrewd administrations that are based on Microsoft Azure encourage associations to produce important information out of it by safely interfacing, overseeing, catching, and changing.
- Applications like Power BI, Office 365, and HD knowledge are utilized to create significant experiences.
- The key highlights are information versatility, readiness, and network.

Amazon Web Services

- AWS offers pay-more only as costs arise model for IoT applications. Amazon Web Services (AWS) licenses the Internet of things (IoT) by empowering administrations, security, and backing.
- It permits quick admittance to wanted to register power through Amazon Elastic Cloud Compute (EC2).
- AWS chains on interest framework for the IoT framework. It gives more stockpiling, process capacity, and worldwide assets.
- It underpins volumes of information and helps in performing huge information examinations.

IBM BlueMix

- IBM offers Bluemix through the stage as an assistance (PaaS) cloud. Coordinated DevOps lets engineers fabricate, test, run, and send just as oversee applications over IBM Bluemix cloud. Bluemix stage runs on the SoftLayer framework.
- The stage is fueled by IBM's after-driving items and administrations: IBM DataPower Gateway, IBM WebSphere Application Server Liberty Core, IBM Informix TimeSeries, IBM MessageSight, Cloudant, and SoftLayer.
- IBM Bluemix stage gives admittance to IoT information and gadgets. It underpins investigation applications, perception dashboards, and portable IoT applications.
- The stage protects API with your application. IBM IoT establishment is the center point where you can set up and deal with your associated gadgets.

Digital Service Cloud

- Advanced help cloud (DSC) is another open IoT stage. This stage enables IoT trailblazers to claim and deal with their clients by associating their items with 1,000,000 different gadgets in an organization.
- It has a fitting and plays dashboard that empowers one to assemble tweaked IoT arrangements.

Zetta

- Zetta is an open-source IoT stage created in Node.js made only for IoT workers that stumble into geo-disseminated workers and on a cloud.
- Zetta is an engineer well disposed, and it runs all over the place: on the cloud, on PCs, and on single-board PCs.
- Zetta has the force and capacity to transform any gadget into an API. The design is advanced for continuous applications that are information exceptional.

Nimbits

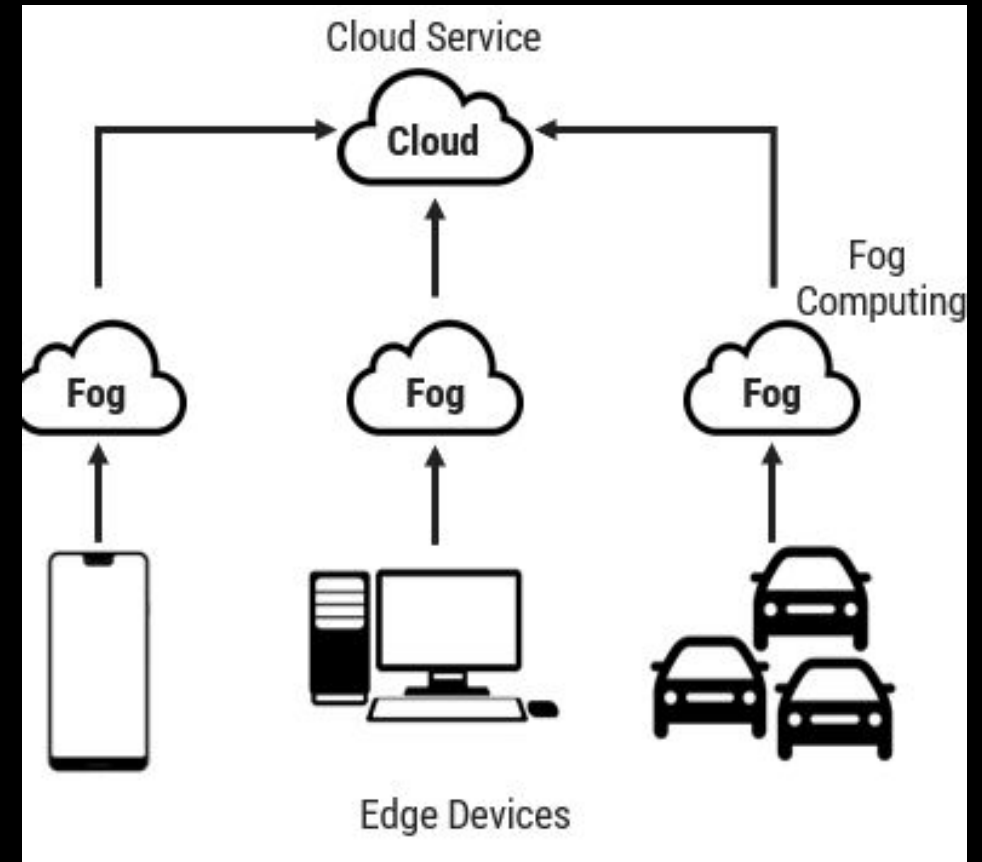
- Nimbits is fundamentally a Platform-as-a-Service (PaaS) that can be downloaded from either Web Server, Raspberry Pi, Google App Engine, or Amazon EC2.
- This stage is utilized for creating both equipment and programming arrangements that interface each other along these lines recovering gigantic volumes of information from actual gadgets and examination.

Emerging Technologies

- **Fog Computing:** Extends cloud computing to the edge, processing data closer to devices for low latency and reduced bandwidth usage.
- Blockchain in IoT security
- Energy Harvesting for IoT Devices (Green IoT)
- 5G connectivity
- LPWAN (Low-Power Wide-Area Networks) Evolution
- Digital Twins
- Generative AI for IoT

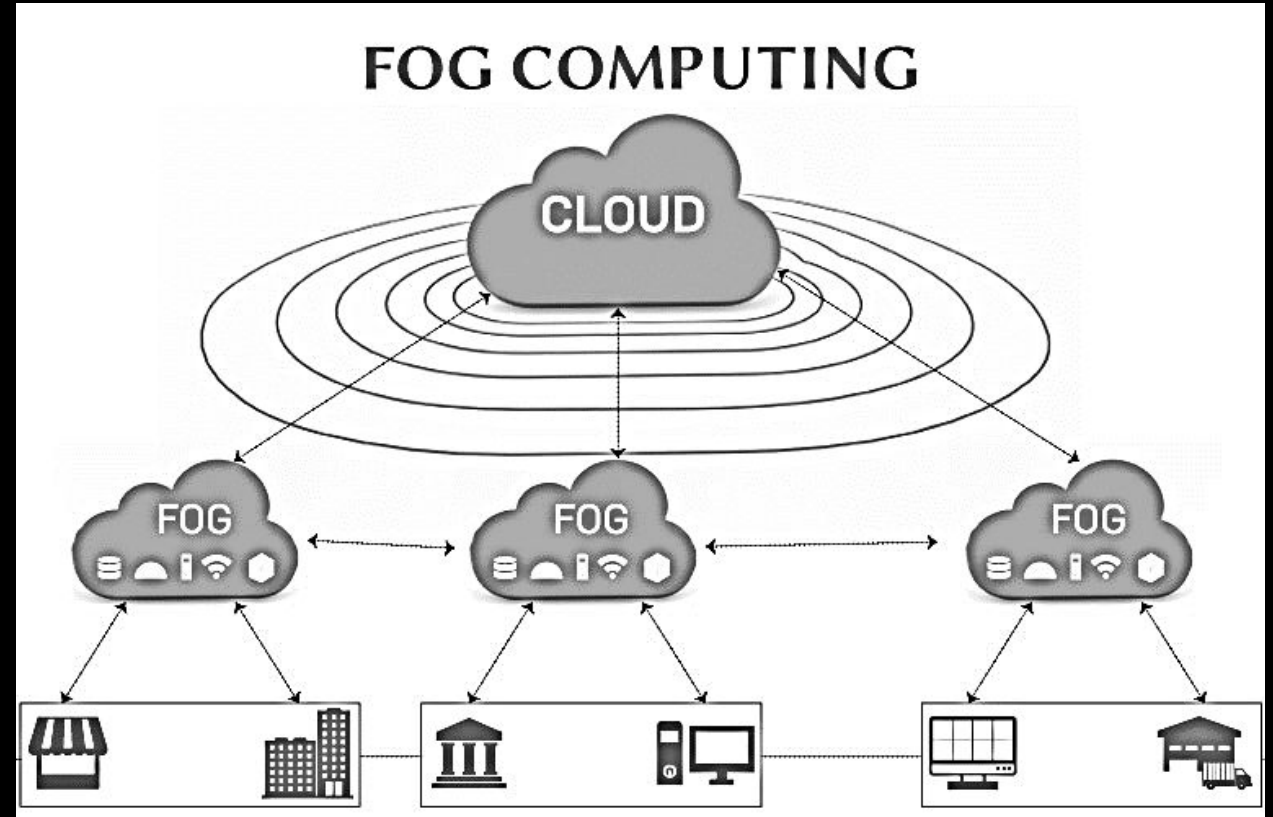
Introduction to Fog Computing

- IoT is all about the data. The factors that affect data are the four Vs - variety, velocity, veracity and volume.
- All IoT applications require instant analysis and action.
- Most of the time, the action would be corrective in nature, it would be business critical.
- In case the data volume is high and it reaches the cloud after some delay.
- So we may lose the opportunity to use the data appropriately.
- In such cases, fog computing serves the solution.



Introduction to Fog Computing

- ❑ The most sensitive data should be analyzed in the area closer to the place where it is generated.
- ❑ With fog computing, it is possible.
- ❑ Using fog computing we can process the data locally and to avoid the trouble by not sending the data to the cloud.
- ❑ Respond much faster because of data is moving locally so data travel is reduced considerably.
- ❑ It thus process the data in milliseconds.

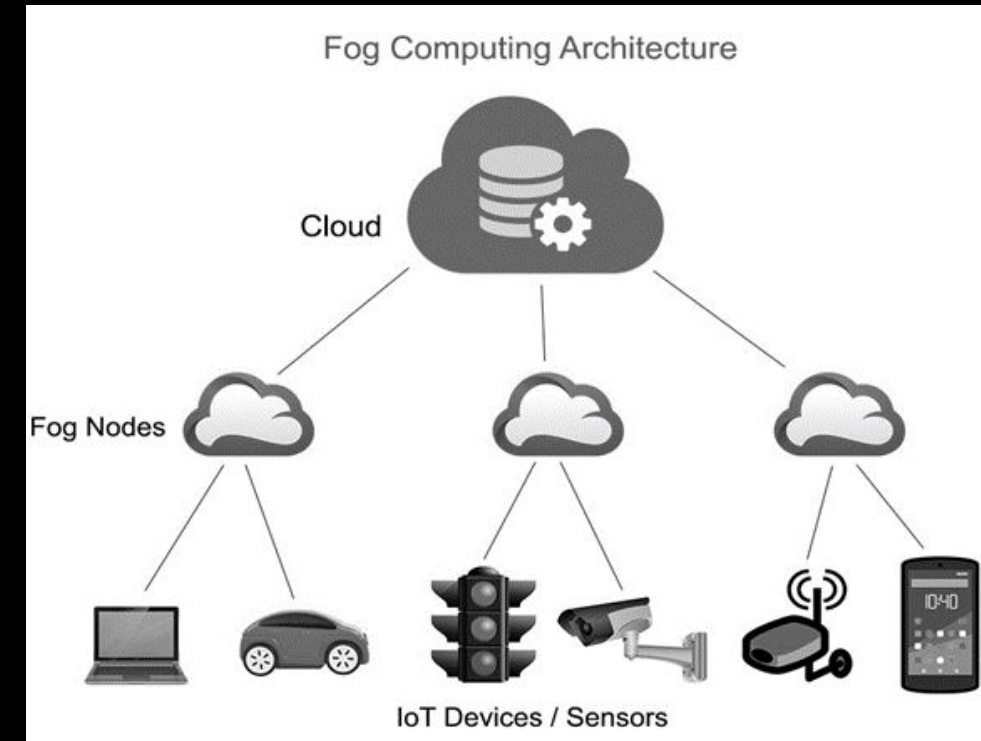


Introduction to Fog Computing

- ❑ Only the required data will be sent to the cloud.
- ❑ This will be based on storage requirements and guidelines.
- ❑ Predictive analytics can also be carried out with the data stored in the cloud.
- ❑ The fog is below cloud, which means it is closer to the elements that generate data.
- ❑ After analysis, the data stored is pushed on to the cloud.
- ❑ Results in increased efficiency and safety both physical and asset safety.
- ❑ Some examples where faster response time is extremely important are factory or manufacturing line, oil and gas tube lines fault analysis, on-flight diagnosis, and healthcare.

Working of Fog Computing

- Sensors/devices generate data transmit it to the middle layer, which is very close the data source.
- These nodes in the middle layer are capable of handling the data.
- This requires minimum power and lesser resources.
- All the data need not go to the cloud at the instant.
- Also, sensitive data gets processed very fast, which results in an instant response.
- Fog is not meant for hefty storage. It is still the cloud that does the task of storing big data.
- Fog is just an intermediary layer for faster data processing, and the faster response time.



Summarize the concepts

Concept of fog nodes

- ❑ It receives the data feed from the sensors, in real-time.
- ❑ Response time is minimal, ideally in milliseconds.
- ❑ Fog computing is transit, where data is stored for a limited time only.
- ❑ Data is then sent to cloud as a summary.
- ❑ It is important to note that not all data goes to the cloud.

Concept of cloud computing platform

- ❑ It receives the data summary from the fog.
- ❑ Data prediction, data analytics, data storage, etc. takes place here.

Benefits of fog computing model

- ❑ Minimal amount of data sent to the cloud.
- ❑ Reduced bandwidth consumption.
- ❑ Reduced data latency.
- ❑ Improved data security. When limited data goes to cloud, it is easier to protect it.
- ❑ Immediate processing of data in real time (this is very much needed in industrial applications).

Edge and Fog Computing

- ❑ Both fog and edge are concerned with the computing capabilities to be executed locally, before passing it to the cloud.
- ❑ Both aim at reducing complete dependency on the cloud to perform computation.
- ❑ Analyzing data and processing it at the cloud is to be avoided.
- ❑ Both these reduce the time delay for making faster decision for real-time applications.
- ❑ The main difference between edge and fog computing is where data processing takes place.
- ❑ Edge computing is the computing carried out at the device itself, where all the sensors are-connected.
- ❑ In fog computing, data processing is moved to the processors that are connected to the local area network (LAN), making it a little farther from the sensors and actuators.
- ❑ Thus, the main difference between edge and fog computing is the distance.